

Taanish Reja

San Diego, CA

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Education

University of California, San Diego (B.S. Computer Science)

2027

GPA: 3.96/4.0

Relevant Coursework

Algorithms and Data Structures, Operating Systems, Parallel Computing (OpenCL/CUDA), Compiler Construction, Distributed Systems, Deep Learning, Reinforcement Learning, Digital Systems Design

Experience

Software Engineering Intern - Ursa Major (TypeScript, React, Three.js, Postgres, Rust)

Jun 2026-Aug 2026

- Building an interactive CAD test-replay viewer that color codes rocket engine components with time-series test data
- Improved internal tool reliability by fixing React state management bugs and adding defensive UI constraints

Instructional Assistant - CSE 120 (Operating Systems) - University of California, San Diego

Mar 2026-Present

- Led weekly office hours with **10+ students**, guiding them through multiprogramming kernel bugs in their OS projects, with **91% of students** reporting highly positive learning outcomes
- Developed **20+ test questions** focused on multiprogramming, synchronization primitives, file systems, and OS abstractions
- Graded **600+ OS assignments**, giving feedback on correctness, race conditions, and low-level systems design

Research Assistant - Mishne Lab

Oct 2023-Dec 2025

- Optimized manifold learning for high-dimensional, low-label-rate datasets, achieving **2x faster** out-of-sample extension and **5% higher** classification accuracy using NumPy, Scikit-learn and CUDA.
- Presented research at the **2024 Learning on Graphs Conference**

Projects

accelerarch.com - Project Management App for Architects (TypeScript, Next.js, Fastify, Postgres)

- Built a project management app unifying scheduling, S3-backed file storage, and client communication, replacing **3 apps** in one architecture firm's workflow
- **Reduced load times 1.5s** by prefetching API calls, caching user auth, and implementing a custom firm/project access model that circumvents Auth0 API limits

bookmarkmanga.com - AI-powered manga tracker (TypeScript, Rust, Next.js, Fastify, Postgres)

- Built a manga tracker that turns user notes and reading activity into semantic graphs, LLM-generated insights, and personalized discussion prompts, supported by a BullMQ message queue and Postgres/pgvector database
- **Cut vectorization costs 65%+** and **runtime 50%** with a custom Rust batch-embedding service

GPU-Rendered UI Library (C++, Swift, Metal)

- Created a lightweight UI library for building desktop applications without a browser runtime
- Built a **retained-mode GPU renderer** with a **DOM tree** and **CSS-inspired styling**, measuring a memory footprint of **35 MB vs. 135 MB** for equivalent Electron code

Climate Forecasting ML Model (Python, PyTorch)

- **Placed 3rd among 83 teams** building models predicting future temperature and precipitation maps from atmospheric data and emissions maps
- **Boosted baseline private score 62.3%** by training **11M-352M** parameter **LSTM and transformer** models with edge-aware losses and squeeze-and-excitation blocks for improved spatial feature learning

Optimizing Compiler for Lisp-Like Language (Rust)

- Built a REPL/JIT/AOT compiler with dead code elimination, constant folding, loop unrolling and inlining, **reducing instruction count 30% vs. baseline**

Peer-to-Peer Coding Collaboration Extension for VS Code (TypeScript, Go)

- Built a collaborative coding extension preserving user edit order under concurrent writes, achieving **sub-110ms edit latency**
- Handled leader failures with Raft election, restoring service **within 1-3s** of leader failure

Skills

Programming Languages: TypeScript/JavaScript, C++, Rust, Python, Go, Java, C, Haskell

Systems: CUDA, OpenCL, Metal, Raft/Paxos, Kafka, Spark

Web & Backend: Next.js, React, Express, Fastify, PostgreSQL, MongoDB, Redis

Machine Learning: PyTorch, NumPy, Scikit-learn, ONNX, OpenCV

Cloud & DevOps: Docker, Kubernetes, AWS (S3, EC2, Lambda), Cloudflare (Domains, SSL, Analytics), GCP